2026 DASU CAPSTONE PROJECT

THUREE



THUREE

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CONTENTS



PROBLEM
&
OUR APPROACH

SYSTEM DESIGN

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EVALUATION

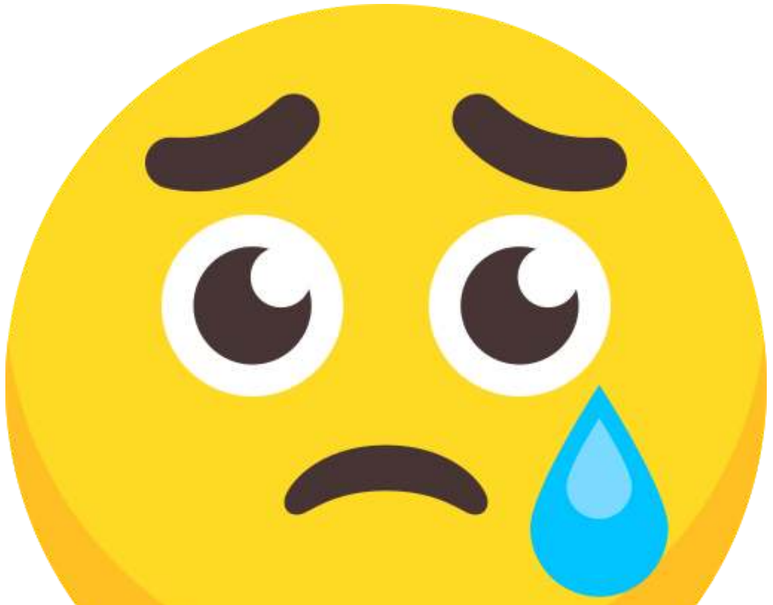
INTRODUCE OUR THUREE





1. PROBLEM

HELP! DASU NEEDS



DASU

Current LLM approaches fail on large files, leaving my answers incomplete :(

FROM DANI



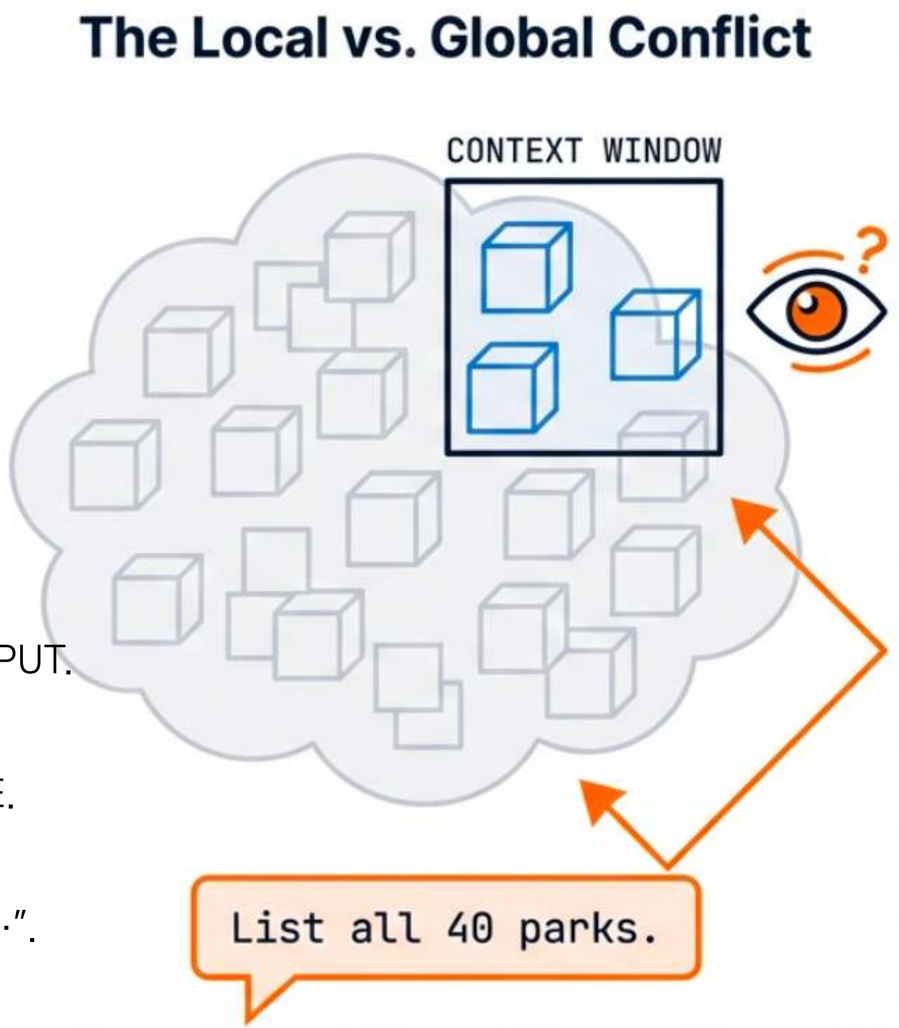
WHY IT DOESN’T WORK?

WHY LLM DOESN'T WORK?

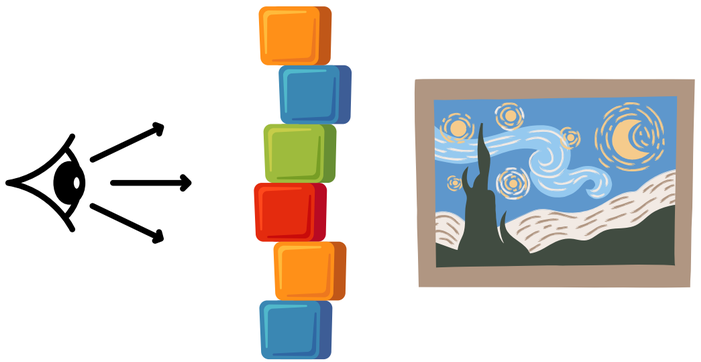
Limitations of Conventional RAG:

Context Fragmentation and Failure on Global Questions

- CONTEXT WINDOW LIMITS
  → 64K TOKEN LIMIT (MISTRAL SMALL 3.2) PREVENTS FULL-DOCUMENT INPUT.
- 'BAG OF CHUNKS' PROBLEM
- SPLITTING DOCUMENTS INTO SIMPLE CHUNKS LOSES GLOBAL STRUCTURE.
- FAILURE CASE:
- CANNOT ANSWER GLOBAL QUESTIONS SUCH AS “LIST ALL…” OR “COUNT…”.



LIMITATION OF TODAY'S RAGS



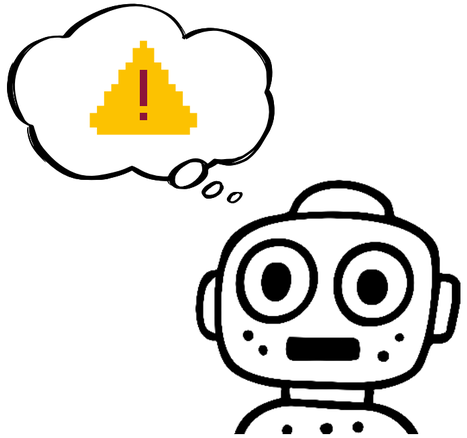
Enumeration Failure

- Blind to the "Big Picture", RAG fails to answer global questions like "List all articles" or "Count titles"
- because it only sees a subset of retrieved fragments.



Document Layout Disruption

- Standard parsers fail to recognize heading levels, treating structured PDFs as a "flat" list of disconnected text.



RAG Hallucinations & Omissions

- Missing key chunks leads to incomplete answers or "hallucinations," where the model wrongly assumes the retrieved part is the whole truth.



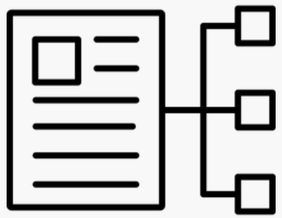
2. OUR APPROACH



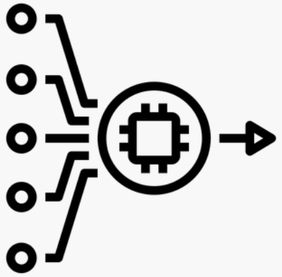
HOW TO SOLVE ...?



Enumeration Failure



Document Layout Disruption



Hallucinations & Omissions

Enumeration Failure

Incorrect or Flattened Document Structure

Static, One-Shot Reasoning

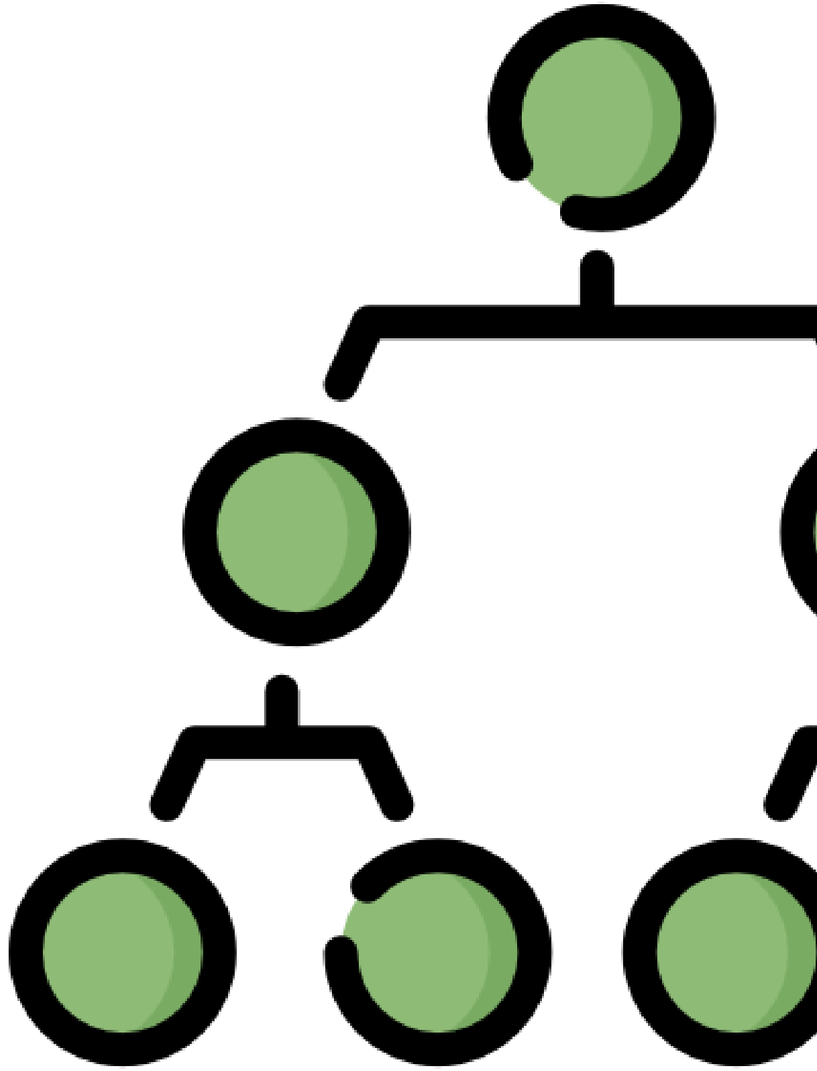
HOW TO SOLVE ...?

Not just a retrieval problem, but a Structure-Aware reasoning problem.

Let’s leverage a Tree Data Structure

Approach this from a data structure perspective

- Structure Preservation
- Exhaustive Coverage
- Hiearacichal Comprehension
- Verifiable Completeness



Let's leverage a Tree Data Structure

Approach this from a data structure perspectiv



Preservation



Coverage



Hiearacichal Comprehension



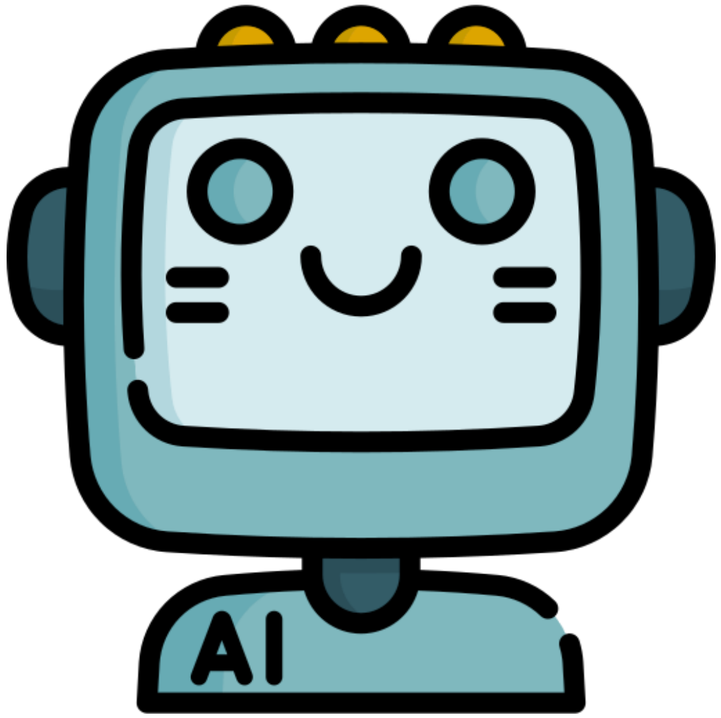
Verifiable Completeness



HOW SHOULD WE CONSTRUCT THE TREE BY REAL DOCUMENT UNITS..?

Understanding Document Structure Like Humans

- Detects **hierarchical patterns** in headings and sections
- Identifies **true document units** and their relationships
- **Reconstructs** an optimized document tree for reasoning



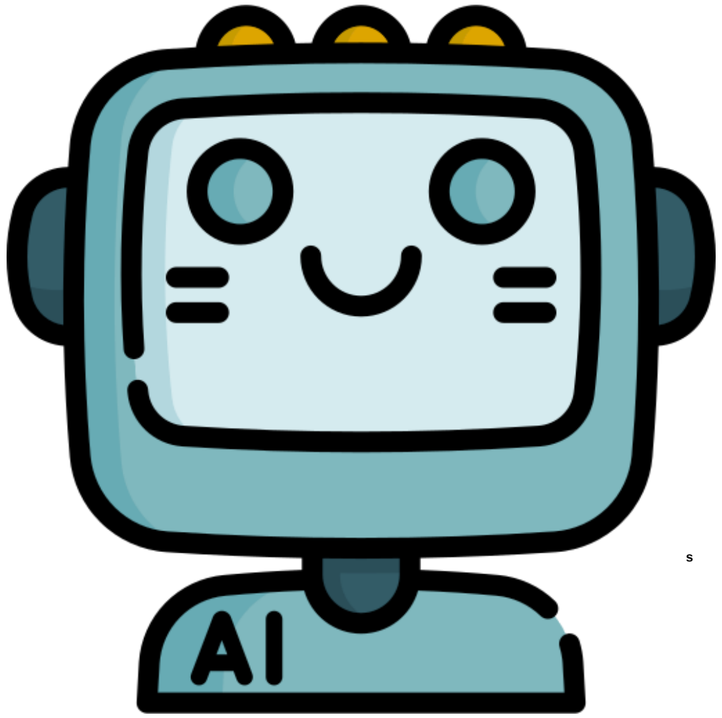
Structure Agent

WE NEED MORE LLM AGENT!



Because…

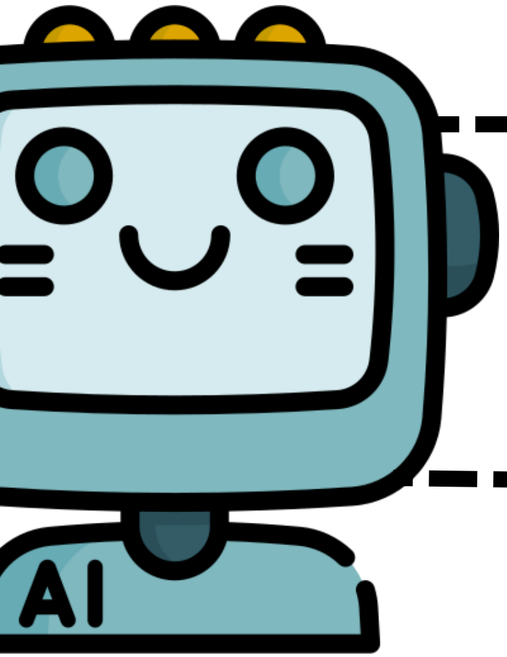
- **Understand intent** & plan reasoning
- Choose and use the **right tools**
- Explore **step-by-step**
- Evaluate and continue reasoning
- Verify for correctness



React Agent

THUREE POINTS

1. Tree Based Storage

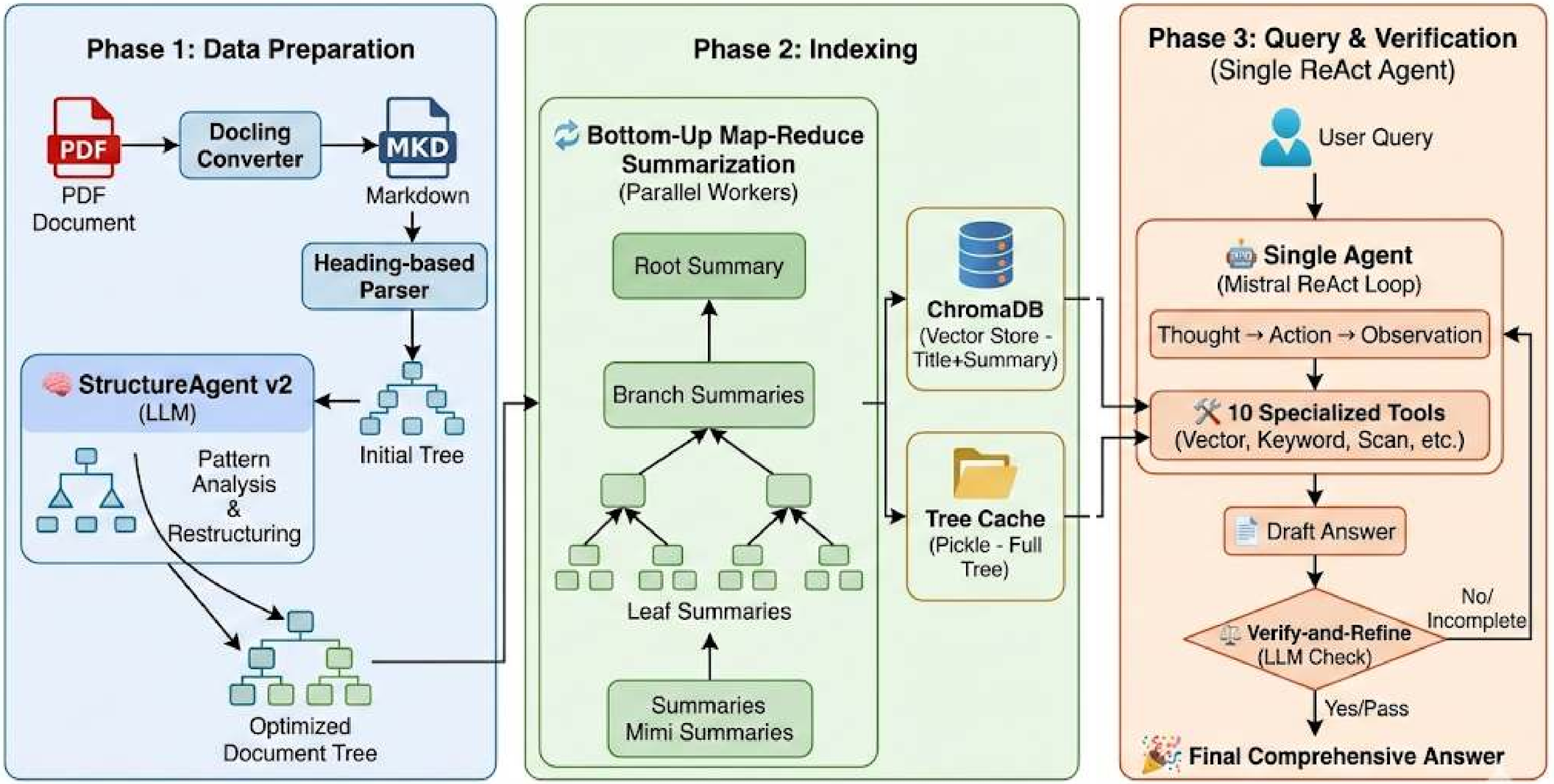


2. Structure Agent

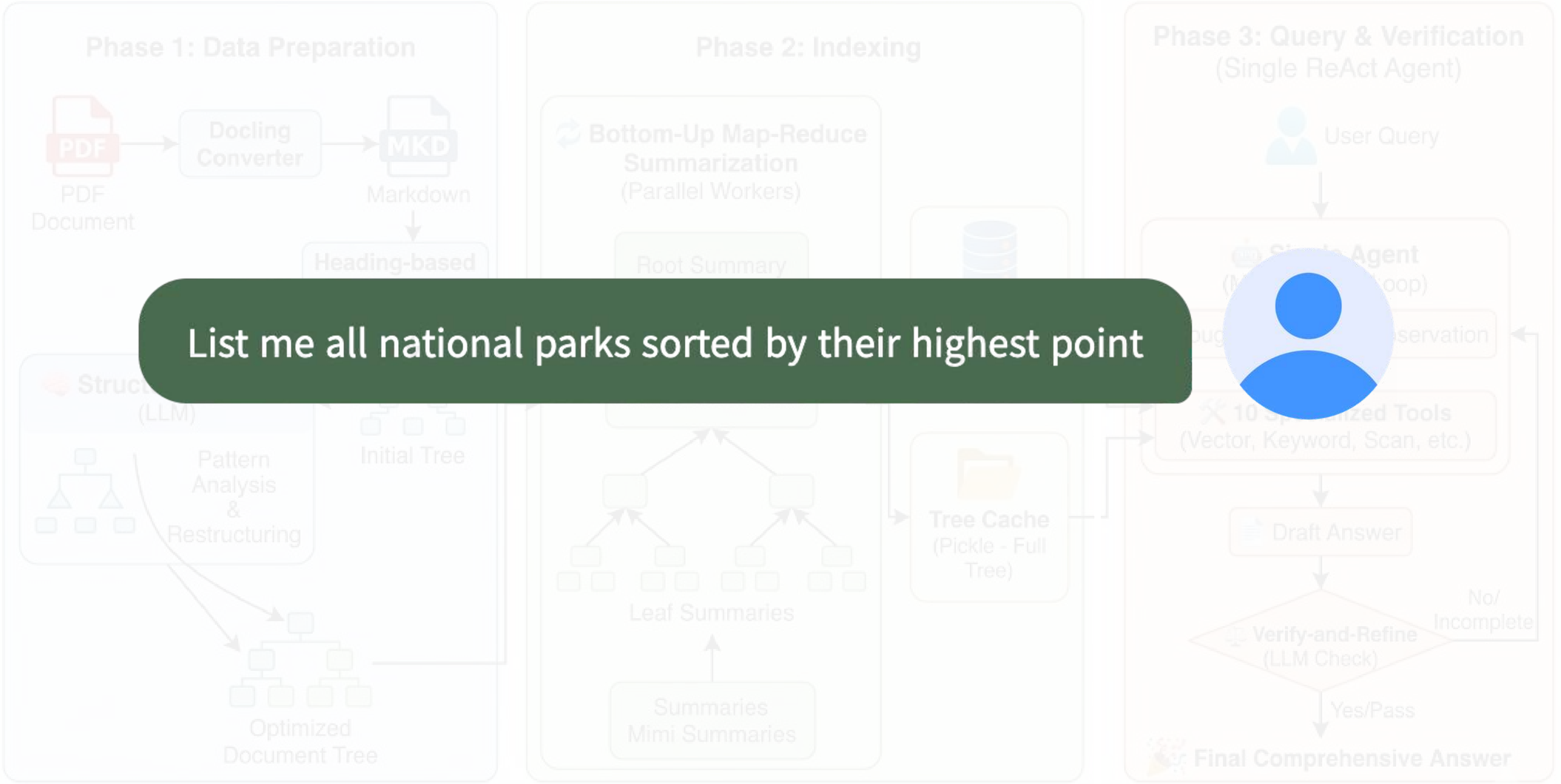
3. React Agent

3. SYSTEM DESIGN

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE

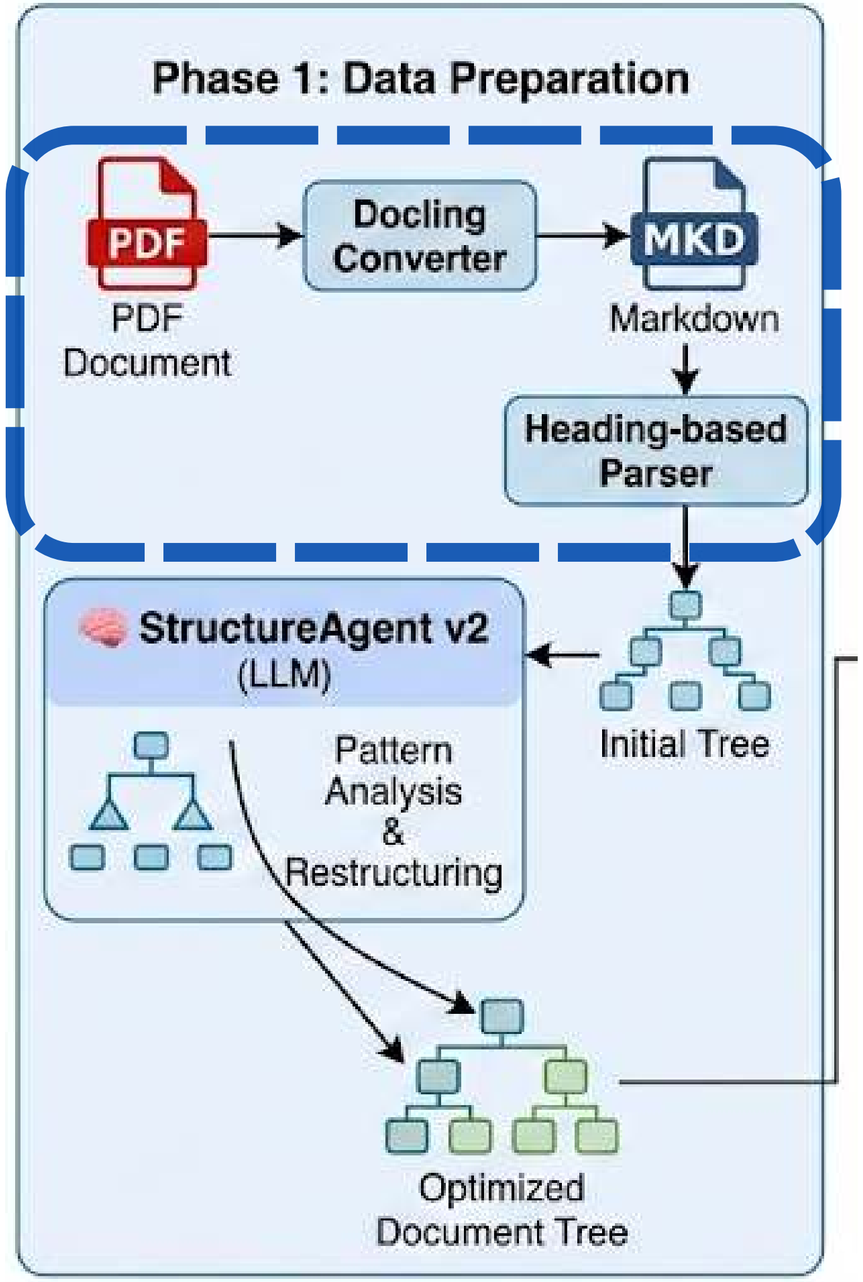


List me all national parks sorted by their highest point



WHY THIS QUESTION IS HARD?

- Nature_Park pdf is **unstructured**
- Requires **ALL entities**, not top-k
- Requires **numeric extraction + sorting**
- Requires **global** document coverage



Phase 1

Building a Tree via **STRUCTURE AGENT**



```
# Abisko National Park
**Country:** Sweden

Beyond the Arctic Circle, Abisko is the nor

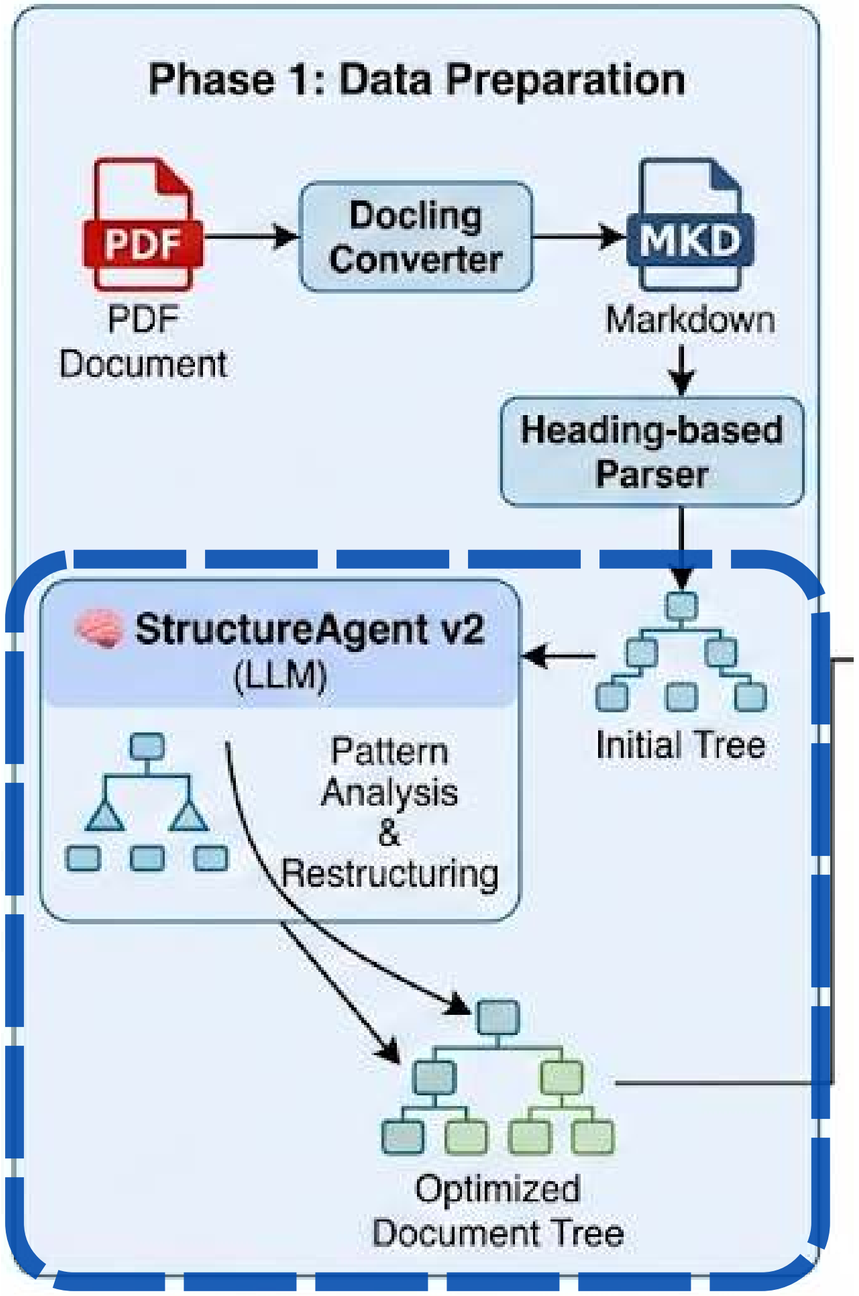
## Park in numbers
- Area covered: 77 sq km
- Highest point: Mount Kebnekaise (2106 m)
- Sámi habitation: 10,000 years

## When to go
- Midnight sun hiking: late May – mid July
- Mosquito season: July – August
- Northern lights skiing: November – March

## Getting there
Abisko is the northern entrance to the 440 k
Nearest airport: Kiruna (2 hours drive).
```

```
Book
├─ Introduction
├─ Abisko National Park
│   ├─ Park in numbers
│   ├─ Wildlife
│   ├─ Hikes
│   └─ Itinerary
```

PDF → Markdown → Heading Tree



Phase 1

Building a Tree via **STRUCTURE AGENT**

WHY STRUCTURE AGENT MATTERS

Without Restructuring

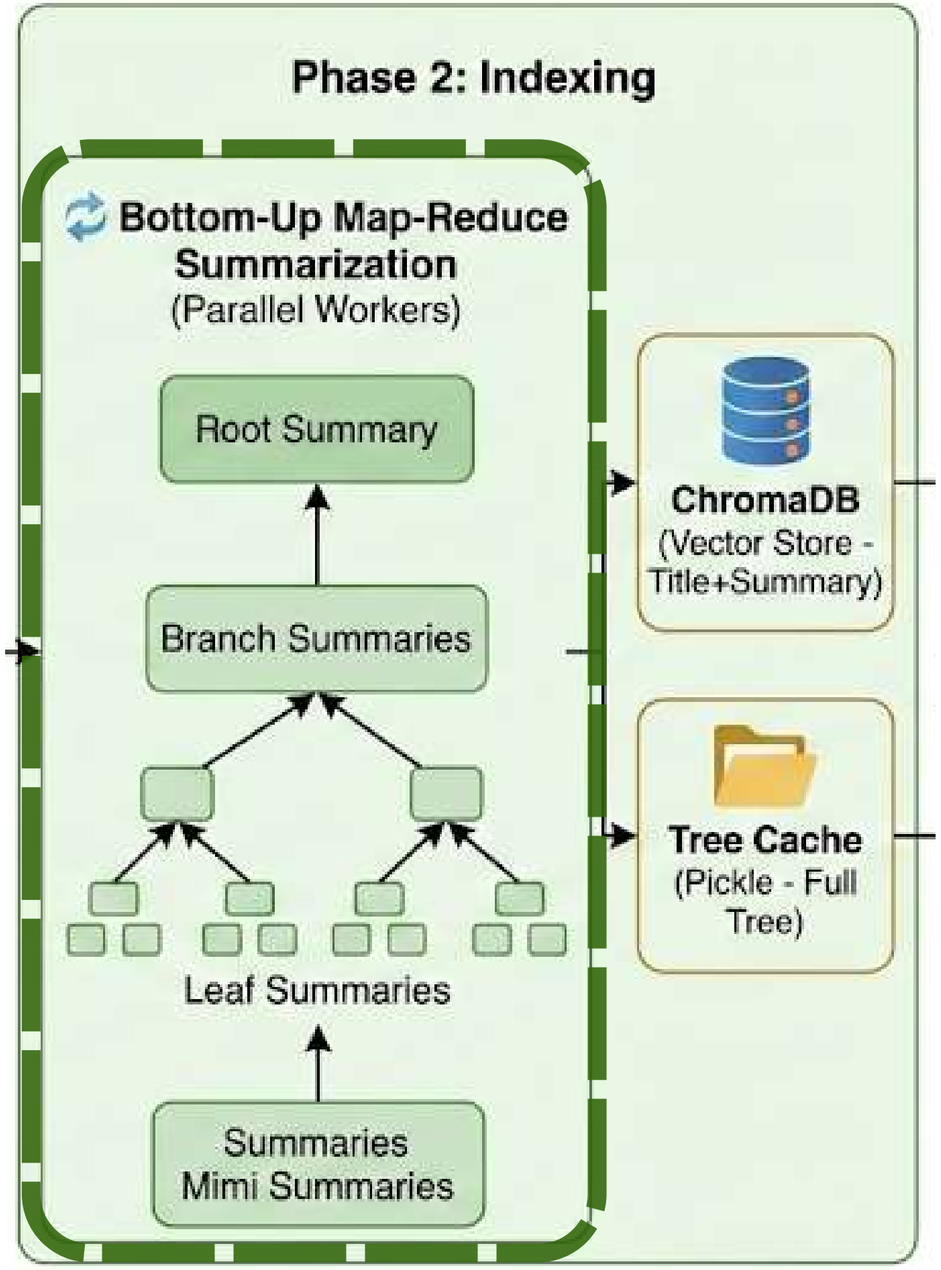
- Heading 1: Introduction
- Heading 1: Wildlife
- Heading 1: Location
- Heading 1: Highest Point
- Heading 1: History

Context lost

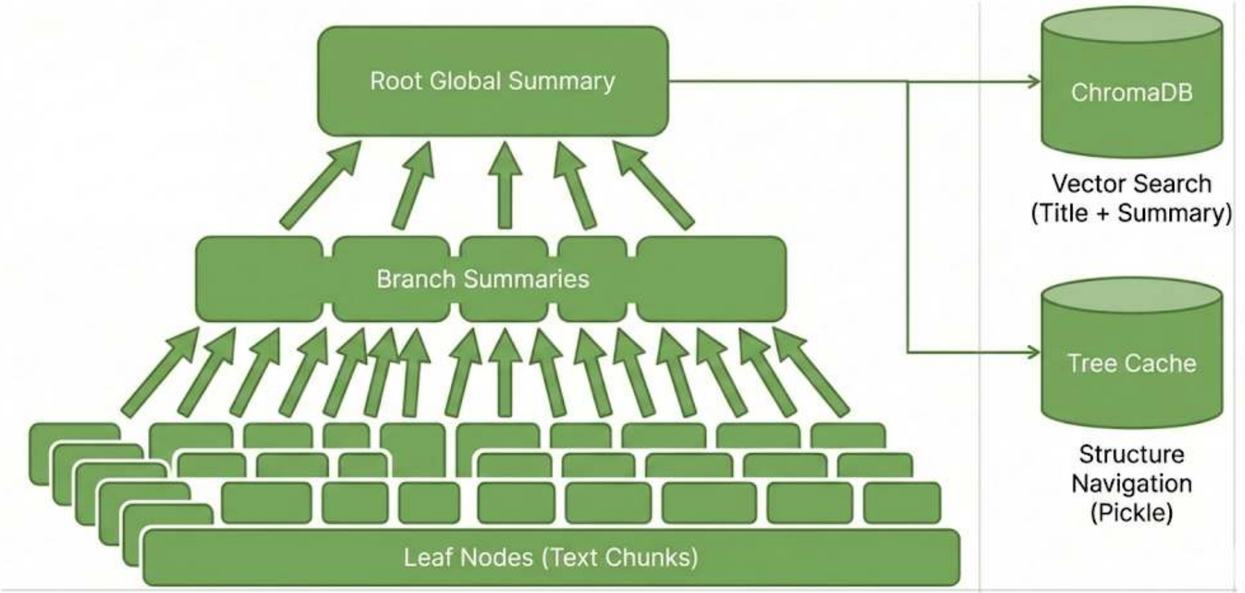
Structure Agent

```
L1 "Abisko National Park"                ← group_name="Abisk
│   meta: {group_name: "Abisko National Park"}
│   summary: "Abisko National Park, established in 190
│   children: 7
│
├─ L2 "Overview"                         ← 5 children (텍스트
│   ├─ L3 [text] "Paragraph 265"   143자
│   ├─ L3 [text] "Paragraph 266"   689자
│   ├─ L3 [text] "Paragraph 267"   995자
│   └─ ...
│
├─ L2 "Toolbox"                          ← 4 children (When
in numbers)
│   ├─ L3 [text] "Merged paragraph" 25자  ← merge_smal
│   ├─ L3 [text] "Paragraph 273"   252자
│   ├─ L3 [section] "Getting there"
│   └─ L3 [section] "Park in numbers"
│
├─ L2 "Stay here..."                     ← 숙소 정보
├─ L2 "Do this!"                         ← 활동 정보
├─ L2 "What to spot..."                  ← 동식물 (wildlife)
├─ L2 "Hike this..."                     ← 하이킹 코스
└─ L2 "Itineraries"                      ← 여행 일정
```

Context Preserved !



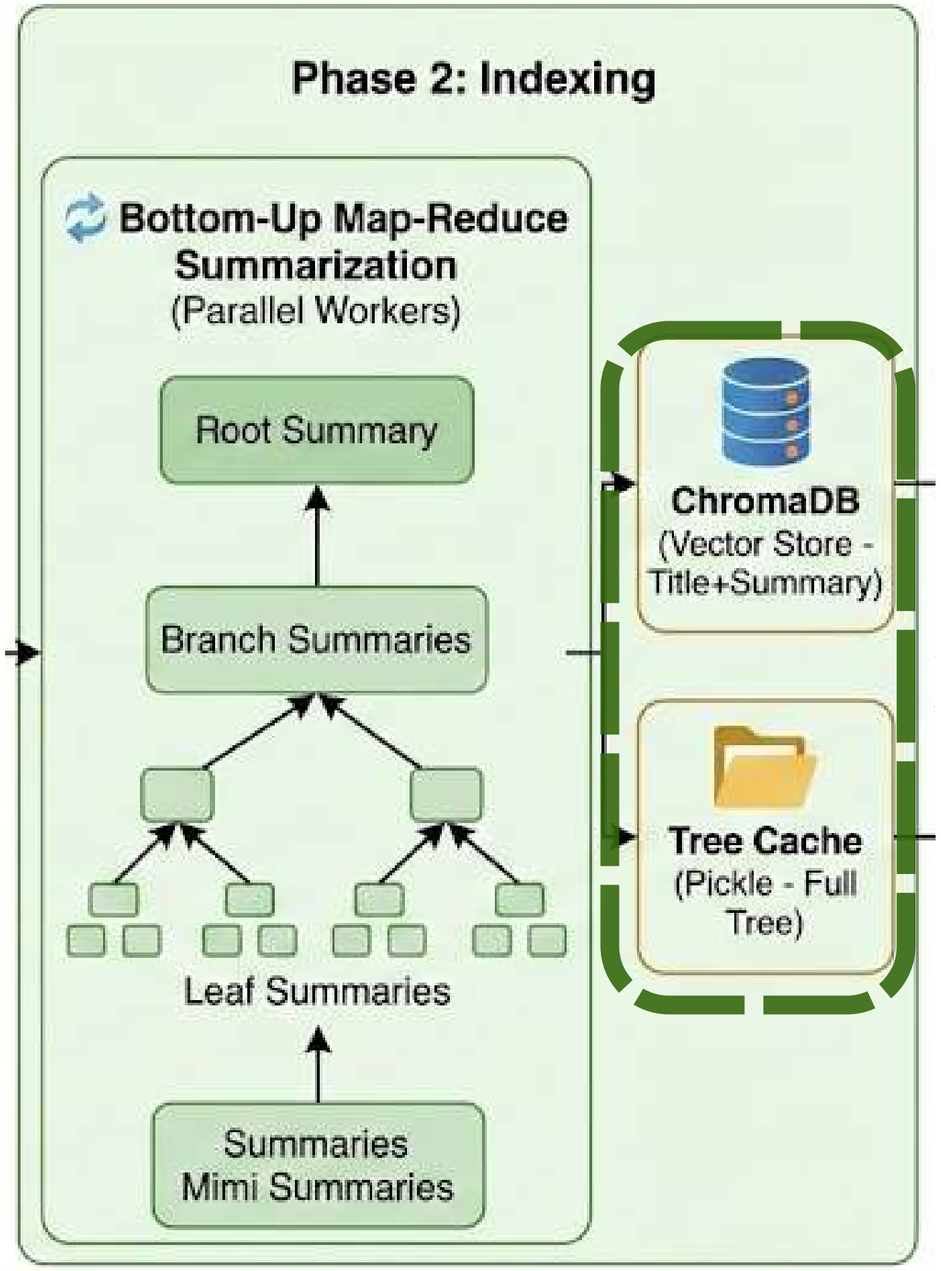
Phase 2
Bottom-Up Indexing



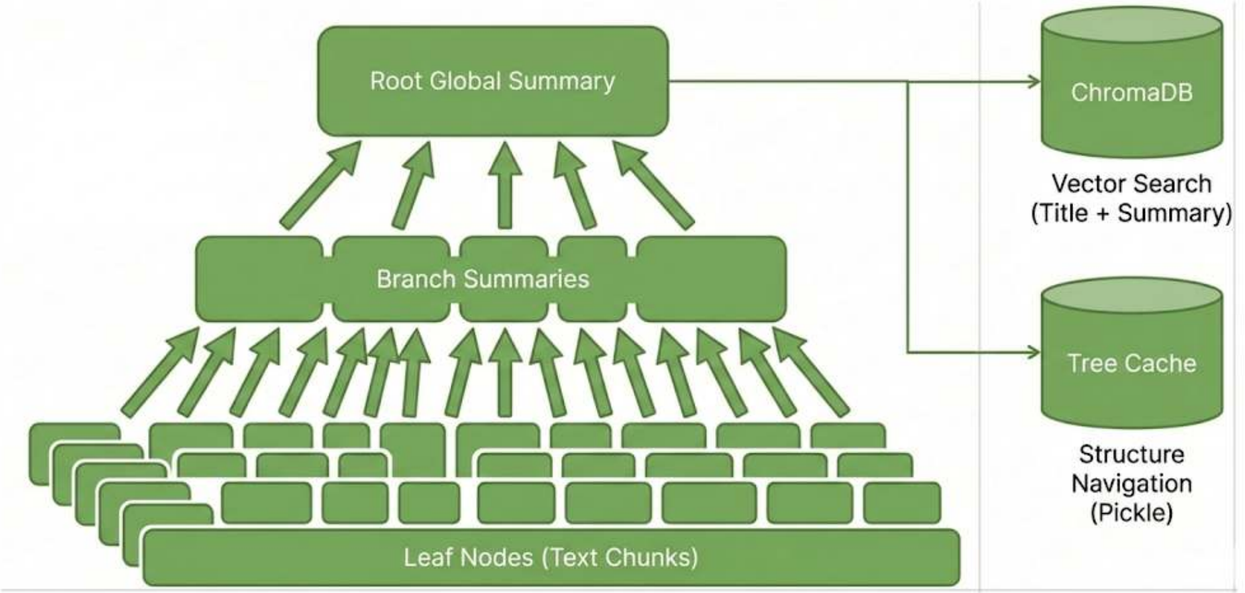
Leaf → Branch→ Root

In Phase 3, the agent must quickly determine

what information each node contains.



Phase 2
Bottom-Up Indexing

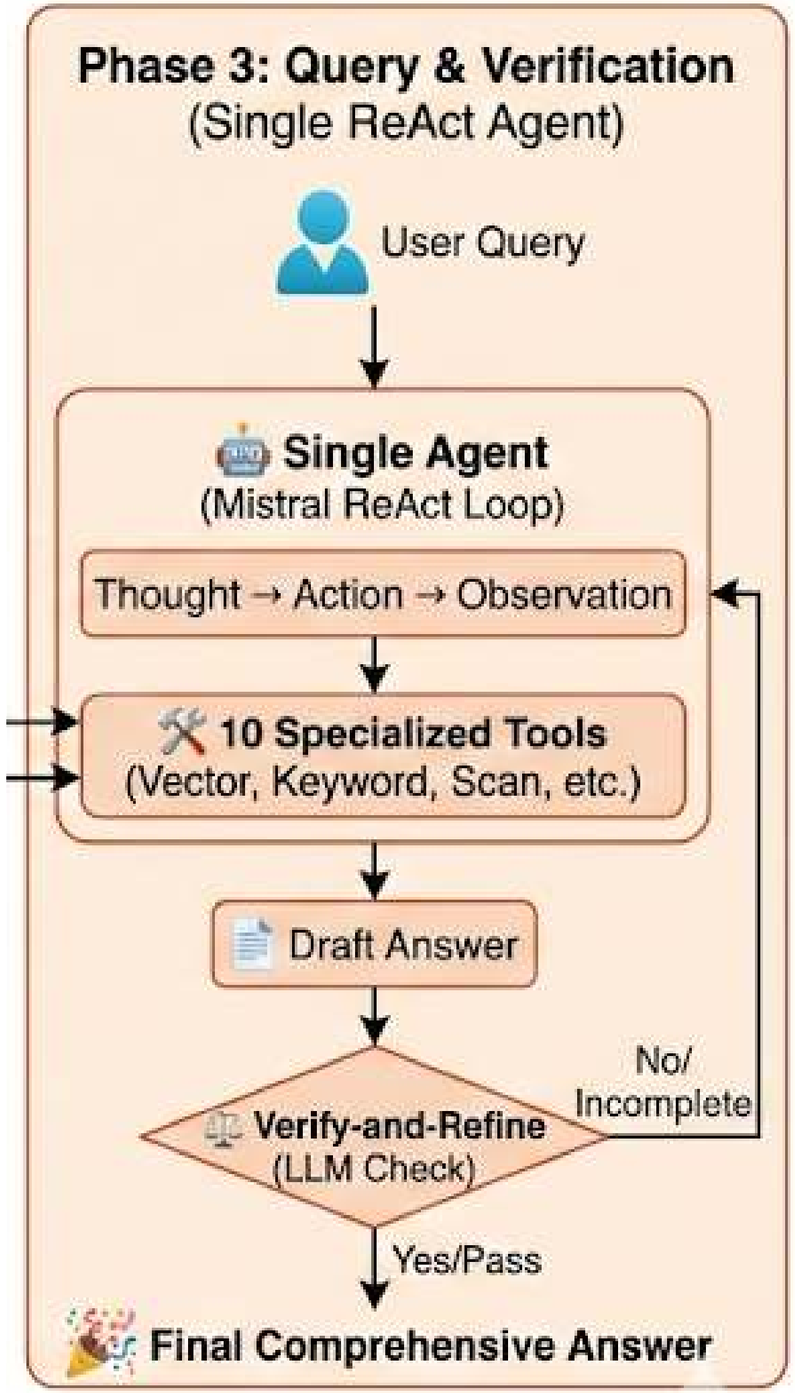


ChromaDB (Vector Store)

- Stores embeddings of each node using its title + summary

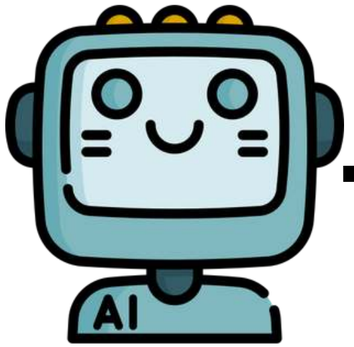
Full Tree Cache

- Serializes and stores the entire tree
  with summaries populated at every node

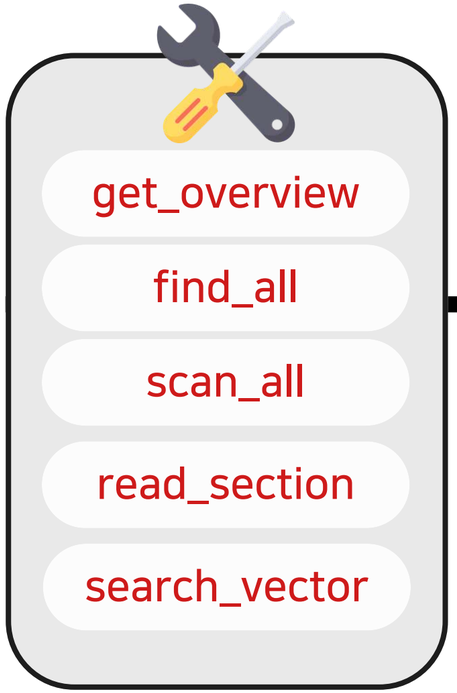


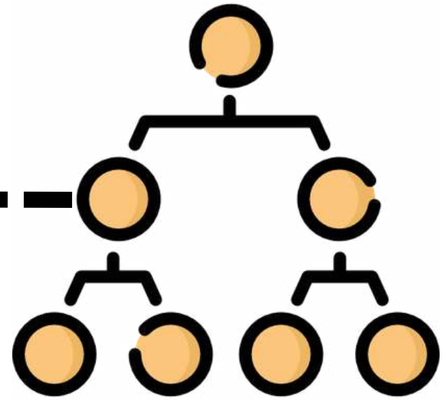
Phase 3
Query & Verification

ReAct Agent



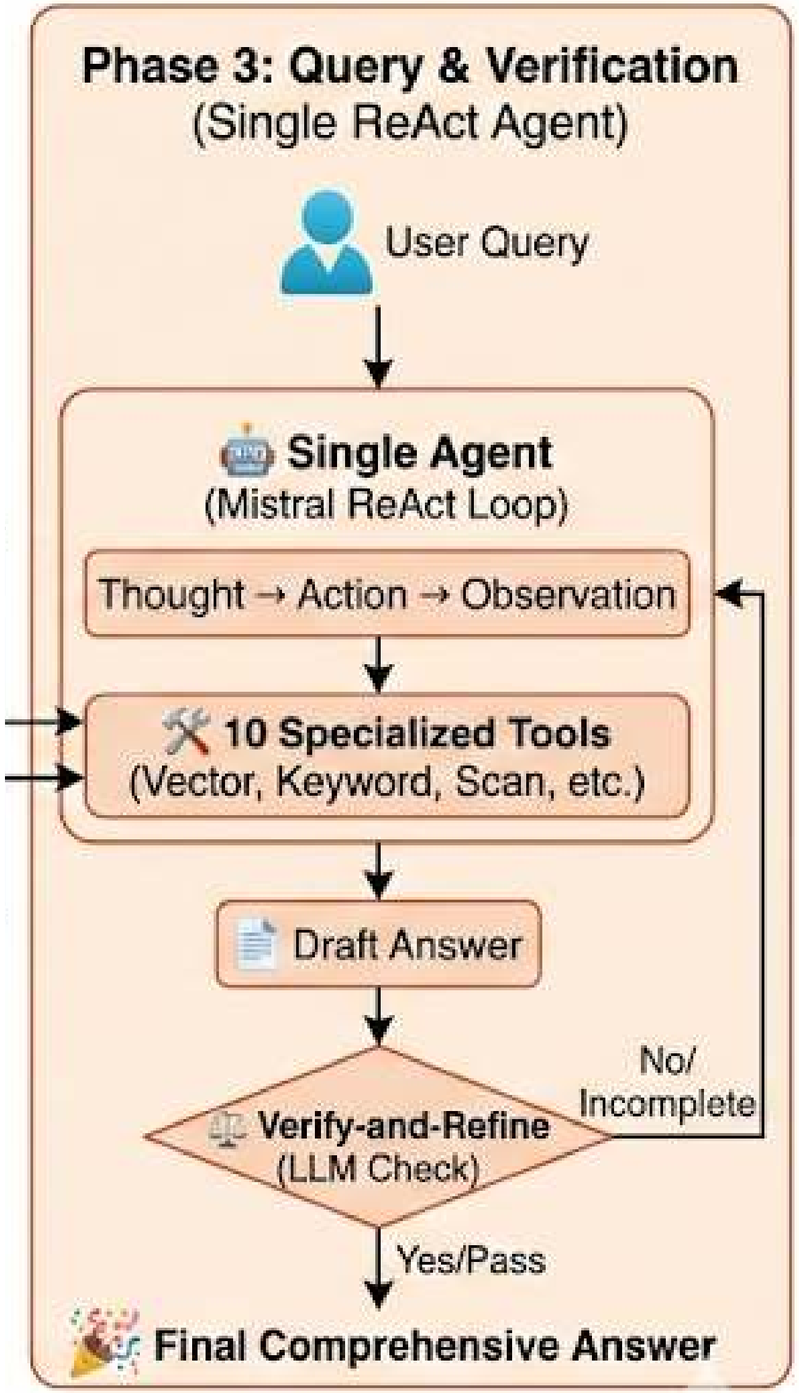
- QUESTION INTENT ANALYSIS
- MULTI-STEP THOUGHT GENERATION
- TOOL PLANNING
- ITERATIVE REASONING (MAX 25 LOOPS)



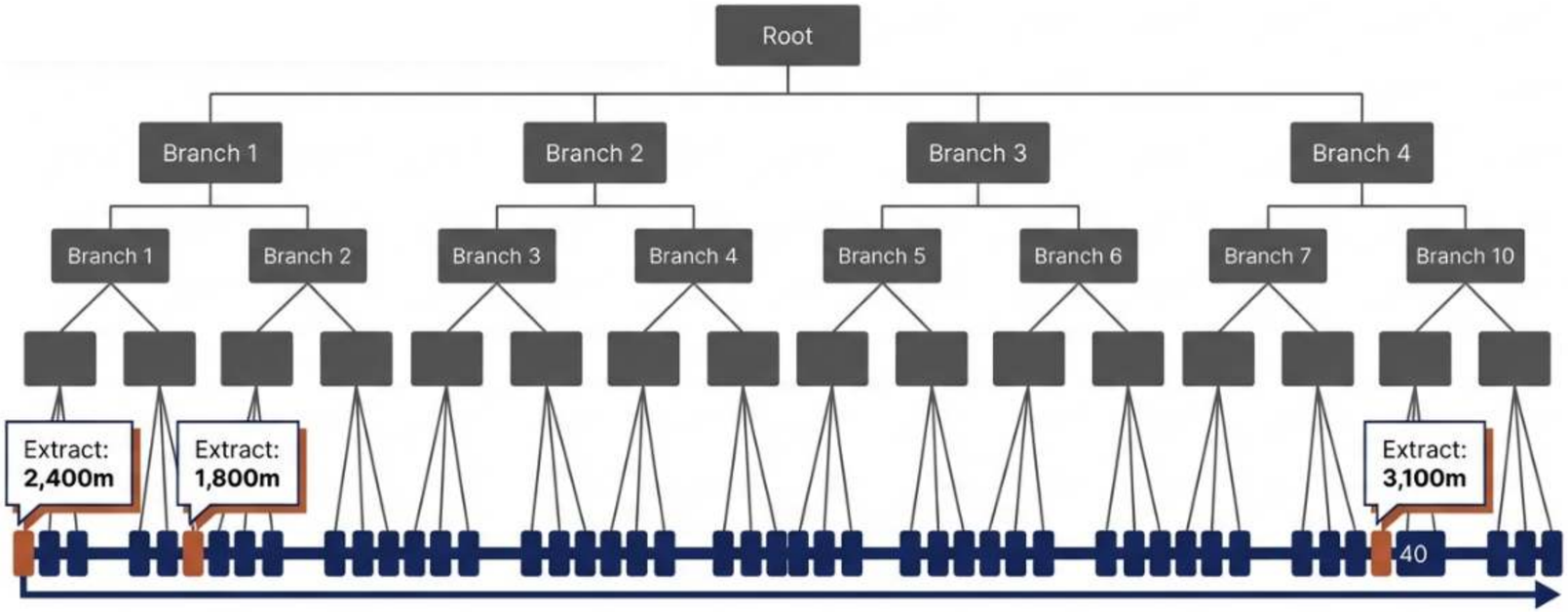


How should the task be performed?

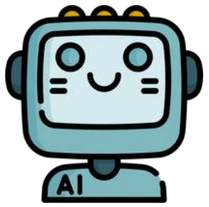
Which tool should be used?



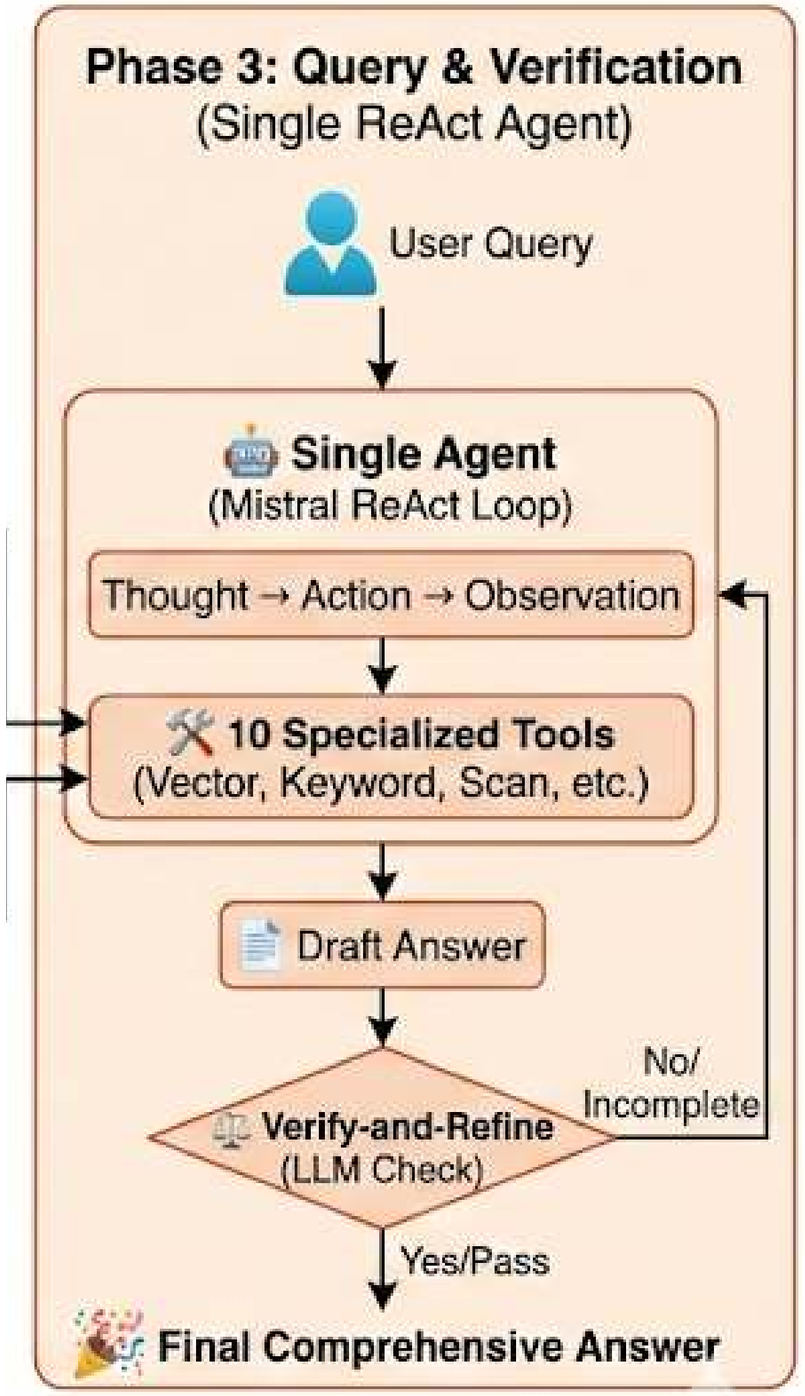
Phase 3
Query & Verification



Bypassing the Context Window: The agent processes leaves in batches, extracting only the height data.

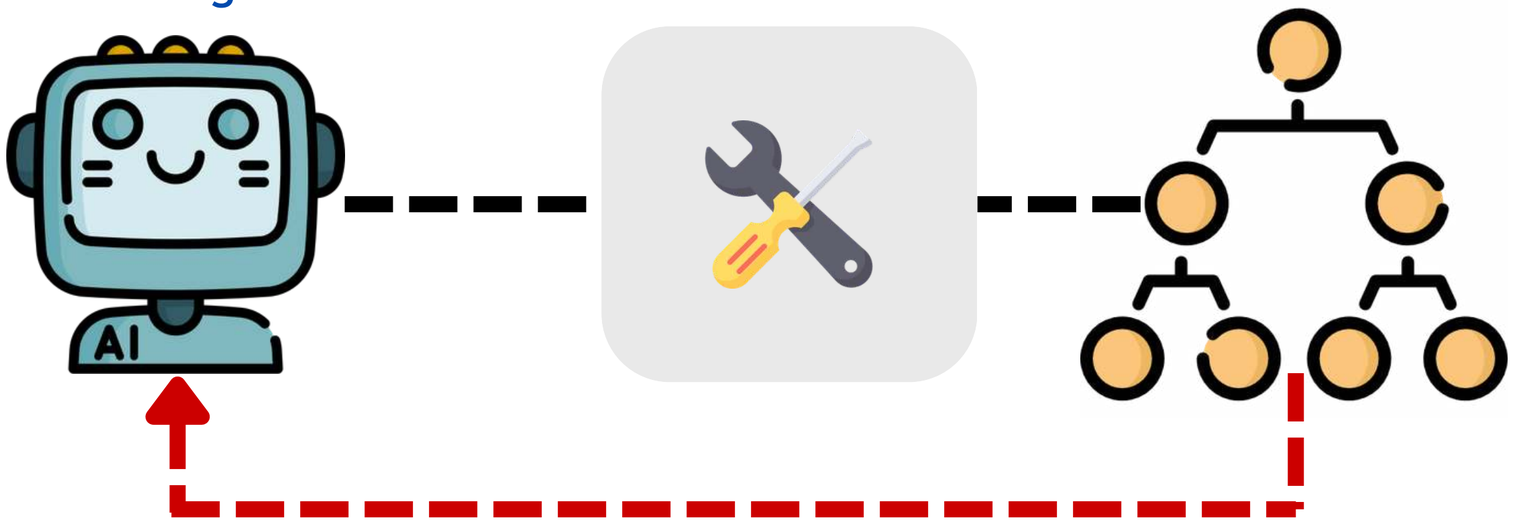


Let’s scan_all use tool!



Phase 3
Query & Verification

ReAct Agent



Verify-and-Refine

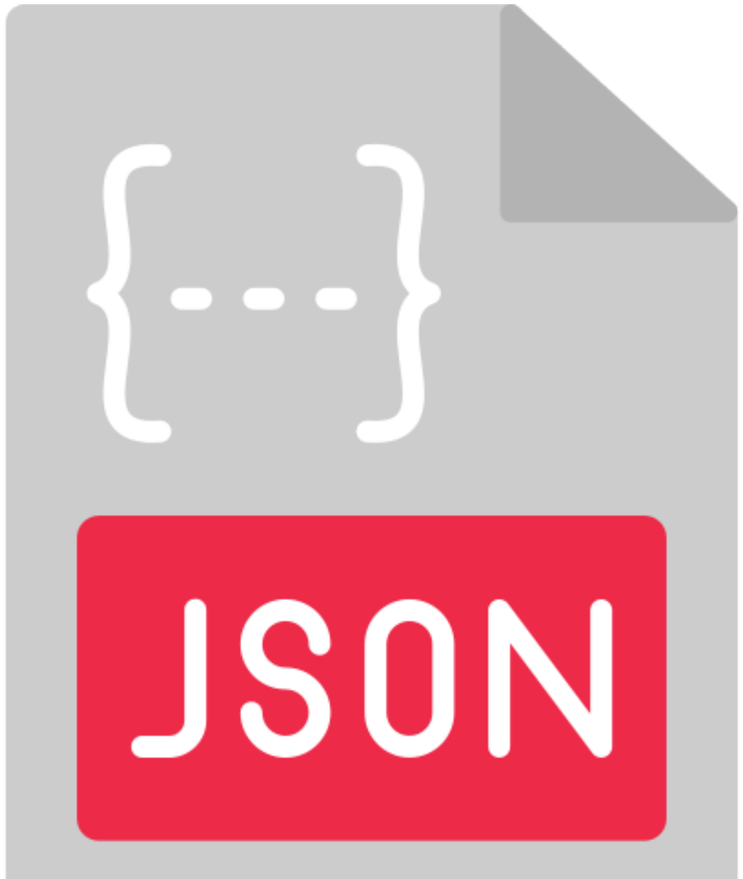
Checking Sorting errors & Source grounding

4. SIMULATION

시연 영상 40초

5. EVALUATION

SO HOW DO WE EVALUATE IT?





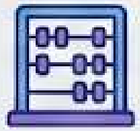
WE BUILT OUR OWN BENCHMARK

FOUR TYPES OF QUESTIONS



(A) Selection

- EXTRACT A SINGLE SPAN OR FACT
- NO AGGREGATION REQUIRED



(B) Enumeration / Counting

- COLLECT ALL ITEMS ACROSS THE DOCUMENT / COUNT THEM
- FULL COVERAGE IS REQUIRED



(C) Aggregation / Summarization

- COMBINE INFORMATION FROM MULTIPLE SECTIONS
- REQUIRES REDUCE-STYLE REASONING



(D) Relational / Multi-hop

- CONNECT INFORMATION FROM DIFFERENT LOCATIONS
- REQUIRES HIERARCHICAL REASONING

EXAMPLE

Dataset Structure

```
Metadata: {
    Metadata
      "Document, (EU Act, Filct Extraction)
       Total Questions)
        "C": Enumeration / Item Collection
        "total Extraction)
    }
       Categories:
        A: Selection
        "C": Enumeration / Item Iclectiount Caculation)",
        "D": Relational/Multi-hop (Connectiong Iffferent Locatio
       }
    },
}
```

Example Questions (Category A)

A01

Question: "What maximum administrrifne uniter Article..?"
Answer: Up to 30,0000,000 EUR or 6%
Answer: 0% a 0% woilwaile annuual turvover.
Source: Article 71(3)

A02

Question: How long must to povider keep EU declaration..."
Answer: Up to ofner woldaide annual ater markets.
Answer: 10 years.
Source: Article 481) / Article 50

ABLATION STUDY

Does Out methods Works?

FI SCORE:
EVIDENCE ACCURACY:
LLM SCORE:







2026 DASU CAPSTONE PROJECT

THUANKU



THUREE

JUYOUNG SEO

SUHWAN JUNG

MINJEONG CHO

HYEONMYOUNG KI

Evaluation Result

dataset
question_id
category
latency
tokens
f1_score
precision
recall
evidence_accuracy
llm_score
reason
answer

adaptive_rag > test_results_full_opt2.csv

```
dataset,question_id,category,latency,tokens,f1_score,precision,recall,evidence_accuracy,llm_score,reason,answ
ai_act_qa.json,A01,A,16.13,124030,0.7018,0.6250,0.8000,True,5.00,The generated answer accurately captures the
ai_act_qa.json,A02,A,10.93,124402,0.5641,0.4583,0.7333,True,4.00,"The answer correctly identifies the 10-year
ai_act_qa.json,A03,A,11.12,119785,0.2857,0.2857,0.2857,True,5.00,"The generated answer correctly identifies t
ai_act_qa.json,A04,A,11.72,123194,0.9367,0.8810,1.0000,True,5.00,"The generated answer matches the ground tru
ai_act_qa.json,A05,A,11.65,124011,0.5405,0.4545,0.6667,True,5.00,The generated answer accurately reflects the
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ai_act_qa.json,A08,A,11.02,121976,0.2941,0.3846,0.2381,True,5.00,"The generated answer correctly identifies A
ai_act_qa.json,A09,A,8.71,121057,0.4000,0.5000,0.3333,True,4.00,"The generated answer correctly identifies th
ai_act_qa.json,A10,A,9.84,121195,0.7586,0.6111,1.0000,True,5.00,The generated answer matches the ground truth
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ai_act_qa.json,B02,B,12.39,122881,0.6465,0.7442,0.5714,True,5.00,"The generated answer accurately identifies
ai_act_qa.json,B03,B,17.24,128222,0.5564,0.4933,0.6379,True,4.00,"The generated answer includes all seven pra
ai_act_qa.json,B04,B,26.69,135567,0.2968,0.1840,0.7667,True,5.00,"The generated answer includes all ten oblig
ai_act_qa.json,B05,B,12.89,126431,0.6889,0.6739,0.7045,True,5.00,"The generated answer accurately identifies
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ai_act_qa.json,B09,B,89.81,168015,0.3125,0.3061,0.3191,False,1.00,"The generated answer incorrectly states th
ai_act_qa.json,B10,B,15.58,121058,0.1053,0.1579,0.0789,False,2.00,"The generated answer lists specific roles
ai_act_qa.json,C01,C,63.52,149368,0.2236,0.1548,0.4022,True,5.00,"The generated answer is highly accurate and
ai_act_qa.json,C02,C,63.89,150689,0.1915,0.1328,0.3434,False,2.00,"The generated answer provides a broad them
ai_act_qa.json,C03,C,30.76,146639,0.3033,0.2883,0.3200,True,3.00,"The generated answer is factually correct a
ai_act_qa.json,C04,C,63.00,150601,0.2426,0.1754,0.3936,True,3.00,"The generated answer provides a broad overv
ai_act_qa.json,C05,C,59.88,149954,0.1813,0.1277,0.3125,False,2.00,"The generated answer provides a broad, hig
ai_act_qa.json,C06,C,63.30,150309,0.2455,0.1653,0.4767,True,2.00,"The generated answer provides a broad summa
ai_act_qa.json,C07,C,68.78,152300,0.2091,0.1403,0.4105,True,3.00,"The generated answer provides a broad summa
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